Rapidly-exploring Random Tree (RRT)

Algorithm 1 RRT

Input: start x_{start}, goal x_{goal}, step size Δ, goal bias p_{goal}
Output: a collision-free path from x_{start} to x_{goal}

$T \leftarrow \{x_{\text{start}}\}$
repeat
  $x_{\text{rand}} \leftarrow$ random point in the whole map ▷ x_{goal} with prob. p_{goal}
  $x_{\text{near}} \leftarrow \textsc{Nearest}(T, x_{\text{rand}})$
  $x_{\text{new}} \leftarrow \textsc{Steer}(x_{\text{near}}, x_{\text{rand}}, \Delta)$
  if not $\textsc{CollisionFree}(x_{\text{near}}, x_{\text{new}})$ **then**
    discard x_{new}; **continue**
  end if
  $T.\textsc{Add}(x_{\text{new}},\ \text{parent} = x_{\text{near}})$
  if $\|x_{\text{new}} - x_{\text{goal}}\| \le \Delta$ **and** $\textsc{CollisionFree}(x_{\text{new}}, x_{\text{goal}})$ **then**
    $T.\textsc{Add}(x_{\text{goal}},\ \text{parent} = x_{\text{new}})$
    return path $x_{\text{start}} \to x_{\text{goal}}$ (follow parents back from x_{goal})
  end if
until path found

Subroutines.

- $\textsc{Nearest}(T, x)$: the tree node closest to x in Euclidean distance.
- $\textsc{Steer}(x_{\text{near}}, x_{\text{rand}}, \Delta) = x_{\text{near}} + \min(\Delta, \|d\|)\,\dfrac{d}{\|d\|}, \quad d = x_{\text{rand}} - x_{\text{near}}$, i.e. move from x_{near} toward x_{rand} by at most Δ.
- $\textsc{CollisionFree}(a, b)$: true if the segment $a \to b$ does not intersect any obstacle (walls inflated by a safety margin).